

PAGE 1 — PATIENT INTAKE & REGISTRATION FORM

Admission: March 15, 2024 | Form INK-2024-039812

Structured intake form covering core PII edge cases: full name vs short name, maiden name cross-reference, dual email addresses, masked SSN, stale prior address, SSN duplicated in both patient and insurance blocks, and employer + job title functioning as quasi-identifiers.

PATIENT INFORMATION		INSURANCE & ID	
Full Legal Name	Sarah Elizabeth Johnson	Photo ID	Texas Driver's License
Preferred Name	Sarah	DL Number	TX-DL-7823041-C (Exp: 09/2026)
Maiden Name	Sarah Elizabeth Nguyen ← name variant	Insurer	Blue Cross Blue Shield TX
Date of Birth	July 22, 1990	Member ID	BCB-TX-00234567
Age / Sex	33 / Female (she/her)	Group Number	GRP-TX-88120
SSN	987-65-4321	Policy Holder SSN	987-65-4321 ← SSN repeated
SSN (masked)	***-**-4321 ← partial form	Secondary Ins.	None
Marital Status	Married		
Home Address	4521 Oak Street, Austin, TX 78701	EMERGENCY CONTACTS	
Prior Address	112 Wickersham Ln, Austin TX 78704 ← stale	Primary	David Allen Johnson (Husband)
Home Phone	(512) 867-5309	Mobile	(512) 867-0092
Mobile Phone	(512) 867-5310	Home	(512) 867-0093
Work Phone	(512) 440-0220 ext. 114	Email	david.johnson84@gmail.com
Email	sarah.e.johnson@gmail.com	Secondary	Carol Anne Nguyen (Mother)
Alt Email	s.johnson@austinisd.org ← work	Phone	(512) 334-8821
Employer	Barton Hills Elementary ← quasi-ID	Address	88 Ridgewood Ln, Round Rock, TX 78664
Job Title	4th grade teacher ← quasi-ID	REFERRING PHYSICIAN	
		Physician	Dr. Patricia Lee, MD
		NPI	1902837465
		Practice	Austin Family Health Associates
		Address	711 W 38th St, Suite 200, Austin TX 78705
		Phone / Fax	(512) 458-3900 / (512) 458-3901

Clerk Notes: Patient also appears in ARC records as 'Sarah Nguyen' (MRN-ARC: ARC-2017-004821, SSN 987-65-4321). Do not confuse with Michael David Johnson (MRN RGH-2023-011820) or Michael James Johnson (MRN RGH-2023-018456) — same DOB, different SSNs.

SECTION 2A — PHYSICIAN INTAKE NOTE

Patient **Sarah Johnson**, a 33-year-old female, presents with 3-week progressive bilateral lower extremity weakness and intermittent blurred vision OD. Referred by PCP **Dr. Patricia Lee** at (512) 458-3900.

History of Present Illness: Ms. Johnson first noticed symptoms climbing stairs at her home at **4521 Oak Street**. On February 22, 2024, a 45-min visual disturbance prompted a call to our after-hours line (512) 300-1002. She works as a 4th grade teacher at Barton Hills Elementary, 4000 Barton Creek Blvd, Austin TX 78735.

Family Hx: Mother **Carol Anne Nguyen** (DOB 09/14/1962): T2DM, HTN. Father **Robert Nguyen** (deceased): CVD.

Allergies: Penicillin (severe), Codeine (moderate)

Plan: Highly suspicious for G35 — RRMS. Ordering MRI Brain/Spine, LP, ophtho referral. All correspondence to **sarah.e.johnson@gmail.com** only — NOT s.johnson@austinsd.org and NOT to Barton Hills Elementary (512) 414-8888. Imaging ordered under DOB **07/22/1990**.

Signed: **Dr. Michael Chen, MD** | NPI: 1234567890 | License TX-MD-00129834
michael.chen@riversidegeneral.org | Pager: (512) 300-1099

SECTION 2B — RADIOLOGY REQUEST EMAIL

From	michael.chen@riversidegeneral.org
To	radiology@riversidegeneral.org
CC	patricia.lee@austinfamilyhealth.com
Subject	Urgent MRI — Johnson, S.E. — DOB 07/22/1990
Date	March 15, 2024 11:04 AM
Patient	Sarah Elizabeth Johnson
DOB	07/22/1990
MRN	RGH-2024-039812
SSN (last 4)	***-**-4321
Insurance ID	BCB-TX-00234567
Ordering MD	Dr. Michael Chen NPI 1234567890
Callback	(512) 300-1099 (pager)

Fax results to Dr. Lee at (512) 458-3901 and upload to patient portal.

SECTION 2C — PATIENT REPLY (portal message)

From: **sarah.e.johnson@gmail.com** | March 19, 2024 7:14 PM

Dr. Chen — husband **David** reachable at (512) 867-0092 or david.johnson84@gmail.com. My mother **Carol Nguyen** ((512) 334-8821, 88 Ridgewood Ln, Round Rock TX 78664) may also call.

Note: I previously went by **Sarah Elizabeth Nguyen** (maiden name). Some ARC records may be under that name. My SSN has always been **987-65-4321**.

— Sarah Johnson | (512) 867-5310

PAGE 3 — FORMAT VARIANT EDGE CASES: PHONE NUMBERS & DATES OF BIRTH

Each row below is a distinct string. str.replace on any one form will not catch the others. All rows in each table refer to the same patient and the same underlying value.

PHONE NUMBER FORMAT VARIANTS — Sarah Johnson, Mobile (512) 867-5310

Source System	Recorded Value	Format	Notes
Paper intake form (scanned)	(512) 867-5310	Standard NANP	Canonical form
Nurse station entry	512-867-5310	Dashes only	No parentheses
Portal auto-import	5128675310	Digits only	No separators
Insurer fax cover	+1 (512) 867-5310	E.164 + NANP	Intl prefix
Legacy HRIS system	512.867.5310	Dot-separated	Legacy format
Referring physician fax	+15128675310	E.164 compact	No spaces
Pharmacy Rx label	512/867-5310	Slash-separated	Unusual separator
Prior erroneous entry	(512) 867-5399	Standard NANP	Wrong — data-entry error
Work phone (separate number)	(512) 440-0220 ext. 114	NANP + extension	Different number entirely

Row 8: '(512) 867-5399' is a data-entry error for the same patient — a completely different string that format-variant rules alone will not catch. Requires separate detection.

DATE OF BIRTH FORMAT VARIANTS — Sarah Johnson, DOB July 22, 1990

Source System	Recorded Value	Format	Notes
Intake form	July 22, 1990	Natural language	Long-form verbose
Insurance portal	07/22/1990	MM/DD/YYYY	US standard
Discharge summary	1990-07-22	ISO 8601	Database / HL7
Legacy ARC record	22-Jul-90	DD-Mon-YY	Legacy clinical sys
Pharmacy Rx label	07/22/90	MM/DD/YY	Short year
Specialist referral	Jul 22 1990	Mon DD YYYY	No comma variant
Nurse note (informal)	born 7/22/90	Prose + date	Embedded in free text
Masked — year only	**/**/1990	Partially masked	Day/month redacted
Quasi-reference	born in 1990	Year only	No day/month — date patterns won't match
Reference without value	DOB on file matches	Reference only	No value — str.replace impossible

Row 9: 'born in 1990' — year-only, no date-format pattern matches. Row 10: 'DOB on file matches' — reference with no value; str.replace has nothing to anchor on.

March 20 — 09:15

Nurse: R. Alvarez, RN

Patient **Sarah Johnson** (DOB 07/22/1990, MRN RGH-2024-039812) called the nurse line at 09:08 from mobile **(512) 867-5310**. Reports worsening right leg weakness. Confirmed address still **4521 Oak Street, Austin TX 78701**. Notified Dr. Chen via pager **(512) 300-1099**.

March 21 — 08:00

Nurse: T. Okafor, RN

MRI results reviewed. Diagnosis confirmed: **G35 — Multiple Sclerosis**. Husband David (DOB: 08/17/1984, SSN **876-54-3210**) requests inclusion in all future discussions. HIPAA auth updated for **David Allen Johnson**; email **david.johnson84@gmail.com** added to portal.

March 22 — 14:00

Nurse: T. Okafor, RN

SW **Jane Kim, LCSW** (ext. 441) met with patient. Employer HR contact: **hr@austinisd.org**, (512) 414-1000 ext. 200. Child **Lily Mae Johnson** (DOB 11/03/2019) — peds: **Dr. Emily Torres, MD**, NPI 8899001122, (512) 448-7900.

INSURANCE CARD — EMBEDDED IMAGE PII

The patient's insurance card below is stored as a raster image scan. PII visible in this image cannot be detected by any text-layer scanner — OCR must run first before detection is possible. This represents a separate, often overlooked detection layer.



Image contains: member name SARAH ELIZABETH JOHNSON — member ID BCB-TX-00234567 — group GRP-TX-88120 — DOB 07/22/1990 — masked SSN ref ***-**-4321 — member services 1-800-521-2227 — MRN RGH-2024-039812 (barcode). All PII exists only as pixel data.

MASKED & PARTIAL PII SPECIMENS

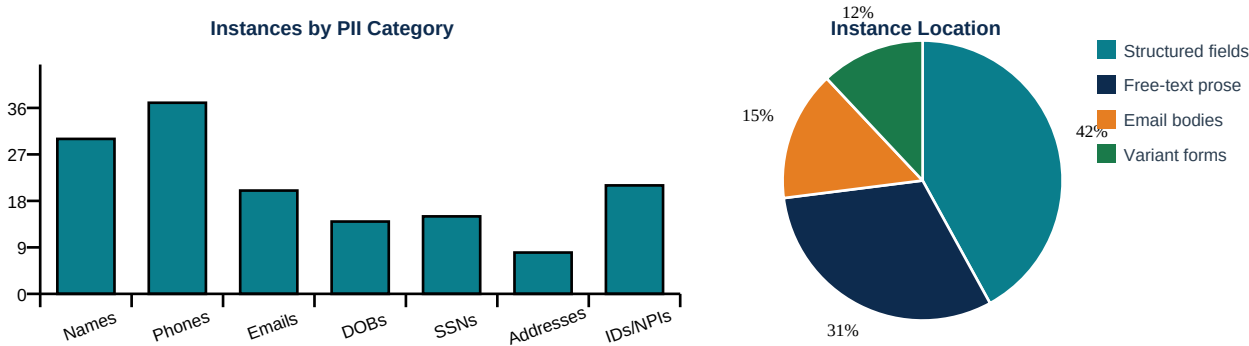
Masked / Partial Form	PII Type	Underlying Value	Detection Challenge
***-**-4321	SSN masked	987-65-4321	Last-4 must match full SSN
ending in 4321	SSN ref	987-65-4321	Prose — no digit pattern
last four digits (4321)	SSN prose	987-65-4321	Embedded in sentence
****4821	Credit card	Card ending 4821	Asterisk-masked number
//1990	DOB yr-only	07/22/1990	Day + month redacted
born in 1990	DOB yr prose	07/22/1990	Year-only, no pattern match
the physician whose NPI ends in ...7890	NPI partial	1234567890	Partial NPI in prose
patient's SSN on file	SSN ref-only	(not stated)	■ No value — unfixable by str.replace
DOB on file matches	DOB ref-only	(not stated)	■ No value — unfixable by str.replace
S.E.J.	Initials	Sarah Elizabeth Johnson	3-letter initial form
Ms. Johnson	Title+last	Sarah Johnson	No first name
Dr. C., our attending neurologist	Name initial	Dr. Michael Chen	Initial form — no pattern match

PAGE 5 — ASSOCIATED RECORDS, LINKAGE & PII ANALYTICS

Multi-person records cross-referencing MRN RGH-2024-039812

11A — PRIOR CLINIC (ARC)		11B — MINOR DEPENDENT		11C — HUSBAND (AUTH REP)	
Filed Under	Sarah Elizabeth Nguyen ← maiden	Name	Lily Mae Johnson ← 3rd party	Name	David Allen Johnson
DOB	07/22/1990	DOB	November 3, 2019	DOB	August 17, 1984
SSN	987-65-4321 ← links records	MRN	RGH-2022-011204	SSN	876-54-3210 ← 3rd-party SSN
MRN-ARC	ARC-2017-004821	Guarantor	Sarah Elizabeth Johnson	Phone	(512) 867-0092
PCP	Dr. Amanda Brooks, MD	Guar. SSN	987-65-4321 ← patient SSN in child record	Email	david.johnson84@gmail.com
Email	amanda.brooks@austinregional.org	Ins. ID	BCB-TX-00234567	Employer	Dell Technologies, Round Rock TX
Phone	(512) 454-3700	Pediatrician	Dr. Emily Torres, MD	Insurance	Aetna PPO — AET-TX-00332198
Old Address	112 Wickersham Ln, Austin TX 78704	Peds NPI	8899001122	Auth Signed	HIPAA 03/21/2024

PII INSTANCE ANALYTICS — Episode RGH-2024-039812 (344 total instances, 143 unique values)



SAME-NAME DISAMBIGUATION — Patients that coexist in this system

Full Name	DOB	SSN	MRN	Note
Michael David Johnson	05/20/1968	111-22-3344	RGH-2023-011820	VP Eng — City of Austin
Michael James Johnson	05/20/1968	222-33-4455	RGH-2023-018456	Staff Eng — same DOB, diff SSN
<i>Sarah Elizabeth Johnson</i>	<i>07/22/1990</i>	<i>987-65-4321</i>	<i>RGH-2024-039812</i>	<i>Current patient — this episode</i>
<i>Sarah Elizabeth Nguyen</i>	<i>07/22/1990</i>	<i>987-65-4321</i>	<i>ARC-2017-004821</i>	<i>Same person — maiden name, prior system</i>

DISCHARGE SUMMARY

Patient	Johnson, Sarah Elizabeth
DOB	1990-07-22 ← ISO format
MRN	RGH-2024-039812
SSN	987-65-4321
Address	4521 Oak Street, Austin, TX 78701
Phone	(512) 867-5310
Insurance	BCBS — BCB-TX-00234567
Admit Dx	Suspected demyelinating episode
Discharge Dx	G35 — Relapsing-Remitting MS

Hospital Course: Methylprednisolone 1g IV daily x 3 days (Mar 20–22). MRI: 4 periventricular lesions. CSF: oligoclonal bands positive. Initiated Interferon beta-1a 30 mcg IM weekly.

Follow-Up:

Neurology	Dr. Michael Chen — Apr 12 (512) 300-1040
Ophthalmology	Dr. Sandra Park — Apr 5 (512) 300-2200
Behavioral	Dr. Lisa Morales — TBD lmorales@austinbehavioral.org
MS Clinic	Apr 20 — first injection training

Faxed to: Dr. Patricia Lee — (512) 458-3901

Signed: **Dr. Michael Chen, MD**
michael.chen@riversidegeneral.org | (512) 300-1040
NPI: 1234567890 | DEA: BC1234523 | License: TX-MD-00129834

SYSTEM METADATA

System IP	192.168.10.45 ← network PII
Session User	jflores@riversidegeneral.org (Janet Flores)
Audit Trail ID	AUD-RGH-2024-039812-0328
Generated	March 28, 2024 Records System v4.2

PHARMACY

Rx Number	RX-2024-082341
Auth Number	AUTH-BCBS-2024-TX-119820
Card Charged	Visa ****4821 ← masked card
Tax ID	74-3829156

SPLIT-LINE PII — LINE BOUNDARY EDGE CASE

In the dictated note below, the physician's email was broken across a line break. The LINE BOUNDARY RULE prevents merging — so neither fragment is a valid email and str.replace on the full address will not find this form:

Line 1 (ends at @)	contact michael.chen@
Line 2 (domain only)	riversidegeneral.org or dial (512) 482-2041
Reconstructed	michael.chen@riversidegeneral.org ← undetectable as-is

QUASI-IDENTIFIERS — RE-IDENTIFICATION BY COMBINATION

These phrases contain no single PII field but combine to re-identify the patient. str.replace cannot address them — semantic detection required:

Quasi-identifier phrase	Re-identifies via
"a 33-year-old teacher in Austin"	Age + occupation + city
"the Johnson residence"	Surname + implicit address
"her school, Barton Hills Elementary"	Employer (indirect)
"her husband's employer in Round Rock"	Relationship + employer city
"Patient with initials S.E.J., DOB 03/14/88"	Initials + DOB combination
"Dr. C., our attending neurologist"	Role + initial (identifiable in context)
"the physician whose NPI ends in ...7890"	Partial NPI reference

FIRST-NAME-ONLY REFERENCES IN PROSE

These appear in free-text and will not be caught by str.replace on full names:

Appearance in prose	Full name behind it	Count
"my husband David can be reached"	David Allen Johnson	6x
"Thank you again, Sarah"	Sarah Elizabeth Johnson	4x
"mother Carol mentioned"	Carol Anne Nguyen	2x
"Dr. Chen reviewed and ordered"	Dr. Michael Chen, MD	10x